

Adopting New Crops

So, when and why do people who are already not only familiar with agriculture, but devoted to an agricultural lifestyle, adopt new crops? A wonderful set of examples comes from historical textual, and occasionally also artistic, evidence of how, when, and sometimes why certain foods were adopted into local agricultural strategies.

A Case Study from the Southeastern United States

Gremillion provides an excellent case study from the southeastern United States in the period following Europeans' arrival in North America. The so-called Protohistoric period of the 16th and 17th centuries, in which Indigenous cultures of the Americas were only indirectly in contact with European colonists, saw new opportunities for those Indigenous societies to adopt select items of European culture—including trade goods and new foods—through trade with other Native American groups who were in direct contact with Europeans. The first food that was adopted in the Southeast was the peach, originally domesticated in China and brought to western Asia and the eastern Mediterranean during the first millennium BCE. It was widely cultivated by the Romans and spread across Europe over the following 1,500 years. It appears to have arrived in the Americas as early as the voyages of Columbus and began to be raised by Spanish colonists in Florida. From there, the peach spread across the Southeast, likely due to its weedy nature: Where a peach pit is planted, a new peach tree is likely to grow!

The peach was quickly adopted by many Indigenous societies of the Southeast, including the Creek and Cherokee, who grew the tree alongside native persimmons and plums close to their villages. Alongside remains of peaches, which are not uncommon to find in paleoethnobotanical study of sites in the Southeast, are rarer finds of watermelon and cow pea (also known as the black-eyed pea, a type of bean first domesticated in Africa). Yet other European crops, including wheat and sugarcane, were ignored by Native inhabitants of the region. Why were these crops adopted and others rejected? It is likely due to how well they fit into Indigenous ways of farming. Peaches, much like the native cultivated fruit trees they were eventually grown with, could be planted and tended with minimal effort and then harvested in the fall alongside other tree fruit. Watermelons, viny plants in the family Cucurbitaceae, grow much like native squashes—which are close relatives in the same family—and could be tended using established and well-known farming techniques. Cow peas are another bean, similar to beans of Mesoamerican origin that had been farmed in the

Southeast for hundreds of years. These new domesticates fit into established farming traditions and agricultural knowledge systems, while wheat, which requires plowing large fields, was completely foreign and un-



Peaches and North American persimmons, watermelons and North American squashes, and cow peas and Mesoamerican runner beans.

Source: Wikimedia Commons: [Peaches](https://commons.wikimedia.org/wiki/File:Prunus_persica_-_Peach_Hungary.jpg)

(https://commons.wikimedia.org/wiki/File:Prunus_persica_-_Peach_Hungary.jpg); [Persimmons](https://commons.wikimedia.org/wiki/File:Diospyros_virginiana-fruit.jpg)

(https://commons.wikimedia.org/wiki/File:Diospyros_virginiana-fruit.jpg); [Watermelons](https://commons.wikimedia.org/wiki/File:Citrullus_lanatus_04.JPG)

(https://commons.wikimedia.org/wiki/File:Citrullus_lanatus_04.JPG); [Squashes](https://commons.wikimedia.org/wiki/File:Cucurbita_pepo_05.09.21_JM_(1).jpg)

([https://commons.wikimedia.org/wiki/File:Cucurbita_pepo_05.09.21_JM_\(1\).jpg](https://commons.wikimedia.org/wiki/File:Cucurbita_pepo_05.09.21_JM_(1).jpg)); [Cow Peas](https://commons.wikimedia.org/wiki/File:Black-)

(<https://commons.wikimedia.org/wiki/File:Black->

[eyed pea pods on plant in Hong Kong.jpg](#); [Runner Beans](#)

(https://commons.wikimedia.org/wiki/File:Phaseolus_coccineus_rip

[e_seedpod_Pronkboon_rjip.jpg](#))

A Case Study from Central Asia

A nexus of trade for thousands of years was Central Asia, the region that stretches from modern Iran to western China, north of the Himalaya mountains and south of Siberia. This dry region includes some of the world's largest deserts and vast areas of dry steppe grassland, as well as several large mountain ranges. Yet this region has always been the best overland route from China to Western Asia and the Mediterranean, and so was used by traders to move high-value commodities across this distance. These routes are commonly known as the "Silk Road," although numerous specific routes linked east and west, without constructed roads, and traders transported much more than just silk, including tea, spices, minerals, and finished goods.



The "Silk Roads" of Central Asia, using first-century CE names of cities and regions.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Silk_Road_in_the_1_centu

[ry_AD_-_en.svg](#))

Across these trade routes, more than just spices and luxury goods was transported. These routes carried ideas, technologies, knowledge, and also plants east or west. Robert Spengler, the author of one reading you completed for this module, is one of the leading archaeologists and paleoethnobotanists studying Central Asia. Spengler's book, *Fruit from the Sands: The Silk Road Origins of the Foods We Eat* (<https://www.ucpress.edu/books/fruit-from-the-sands/paper>), summarizes his work and that of many other scholars. It brings together historical and archaeological evidence to reconstruct the timing and paths of plant dispersals across Central Asia, aiming also to explain why these foods became key parts of the cuisine (or not!) in the regions into which they were introduced. You read the chapter on various fruits and nuts that were transported as part of the long history of exchange through Central Asia. Let's focus here on two: peaches and melons.

We have already seen that peaches were a key species readily adapted by Native Americans in the U.S. Southeast, but the Europeans who introduced the peach gained it from the legacy of its spread during the Roman Empire (more on this next); and the Romans, in turn, received peaches through Central Asian trade networks from their location of origin and domestication: China. Although wild species of peach are found across western and northern China, the likely location of origin of the domesticated peach is the lower Yangtze River Valley in the area of modern Zhejiang Province. This region has the oldest and most abundant paleoethnobotanical evidence of peach consumption, and the peach pits recovered archaeologically match modern peaches closely in size and shape. It is likely that peaches began to undergo the domestication process in this region by around 5500 BCE and reached modern peach sizes by 3300 BCE. Peaches not only reached the Mediterranean but also were adopted along the Silk Road; there is archaeological evidence of cultivation in western China (Gansu and Xinjiang) and Uzbekistan, and historical accounts of their widespread cultivation where sufficient water was available.

A return journey was made by the melon, a species likely domesticated in Southwest Asia (probably in modern Iraq, Iran, or Turkey) and traded from west to east along the Central Asian trade corridors. This movement took place later than the westward exchange of peaches, with melons not appearing in China until the middle of the first millennium CE—the same time as their earliest archaeological discovery in western Uzbekistan. Not only were melons quickly cultivated along the Silk Road, but local varieties were quickly developed and widely adopted. Across Central Asia today, there is no fruit more closely tied to specific areas, nor as culturally valued, as melons during peak season. These melons come in a vast array of colors, sizes, shapes, and patterns, and at times, it seems that each vill-





A farmer shows his own prized variety of melon in western Uzbekistan (top); a produce market in Samarkand, Uzbekistan in 1911 (middle); and Chinese heirloom melons known as Hami in a Xinjiang market (bottom).

Source: Dr. Marston; [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Prokudin-Gorskii-15.jpg)

(<https://commons.wikimedia.org/wiki/File:Prokudin-Gorskii-15.jpg>);

[Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Xinjiang,_China_(27637339323).jpg)

([https://commons.wikimedia.org/wiki/File:Xinjiang,_China_\(27637339323\).jpg](https://commons.wikimedia.org/wiki/File:Xinjiang,_China_(27637339323).jpg))

Why did peaches and melons find ready acceptance, both at the opposite end of the Silk Road and within communities along it? Both are easy crops to farm where sufficient water is available, and such areas were also where most farming populations settled. Both resemble other crops traditionally grown in these regions—including, in the case of peaches, native plums and cherries, present across Eurasia. Melons are all members of the Cucurbitaceae family and grow similarly to other gourds and melons already established in Chinese agricultural practices. The know-how needed to grow both fruits existed. But why and how did they enter culinary practices across the

region? What made these fruits accepted and valued foods? We can again point to similarities with other fruits, but another, simpler answer is that they are both sweet and easy to prepare. Both fruits readily appeal to humans' innate preferences and evolutionary history as omnivores who value highly sweet fruits. Perhaps it would have been more of a surprise if these fruits were rejected rather than accepted into local cuisine!